

$$a(x,y)u_{xx} + b(x,y)u_{xy} + c(x,y)u_{yy} = f(x, u_x, u_y, x, y)$$

• Discriminant: $\Delta = b^2 - 4ac$

(i) $\Delta > 0$: Hyperbolic:

• 2 BC's

• 2 IC's

ex: Wave equation: $u_{tt} = u_{xx}$

$$a(x,y) = 1$$

$$c(x,y) = -1$$

$$b(x,y) = 0$$

$$\Delta = 0 - 4(1)(-1) = 4 > 0$$

(ii) $\Delta = 0$ Parabolic:

• 1 IC

• 2 BC

ex: Heat equation: $u_t = f$, $a=1$, $b=c=0$

$$\Delta = 0 - 0 = 0$$

(iii) $\Delta < 0$: elliptic:

* Only requires BC's

ex: Laplace Equation: $u_{xx} + u_{yy} = 0$

$$a=c=1, b=0$$

$$\Delta = -4(1)(1) = -4 < 0$$

* A Homogeneous B.C. is one st. if u solves the BC, then all (for all α) also satisfies the B.C. (suppose ϕ is a function of x)

ex: $\phi(0) = 0$ is a HBC, since $\alpha\phi(0) = \alpha \cdot 0 = 0$

$\phi(\pi) = 1$ is not a HBC, since $\alpha\phi(\pi) = \alpha \cdot 1 = \alpha \neq 1$

* Superposition also holds for linear PDE's $\forall \alpha$

$\Rightarrow \phi_1$ and ϕ_2 are linear sol^{ns} then

$\alpha\phi_1 + \beta\phi_2$ also solves it

Boundary Value Problem: $X(x) : \begin{cases} X'' = -\lambda X \\ X(0) = 0, X(L) = 0 \end{cases}$

$$0 \quad L$$

• λ : undetermined, $X(x) = e^{rx} \Rightarrow r^2 = -\lambda, r = \pm \sqrt{-\lambda}$

Case 1: $\lambda > 0, r = \pm i\sqrt{\lambda}$

$$X(x) = A \cos(\sqrt{\lambda}x) + B \sin(\sqrt{\lambda}x)$$

$$X(0) = 0 = A$$

$$X(x) = B \sin(\sqrt{\lambda}x), \quad X(L) = 0 \Rightarrow$$

$$\sin(\sqrt{\lambda}L) = 0 \quad \sqrt{\lambda}L = n\pi$$

$$\sqrt{\lambda} = \frac{n\pi}{L} \Rightarrow \lambda_n = \frac{n^2\pi^2}{L^2}, \quad n=1, 2, \dots$$

$$X_n(x) = B_n \sin\left(\frac{n\pi}{L}x\right) \quad (\text{Eigen vectors})$$

$$\lambda_n = \frac{n^2\pi^2}{L^2}, \quad n=1, 2, \dots \quad (\text{Eigenvalues})$$

(ii) $\lambda = 0$

$$X(x) = Ax + B$$

$$X(x) = 0$$

$$X(0) = B = 0$$

$$X(L) = AL = 0 \Rightarrow A = 0$$

Trivial solution

(iii) $\lambda < 0$

$$\hat{\lambda} = -\lambda$$

$$X(x) = A \sinh(\sqrt{\hat{\lambda}}x) + B \cosh(\sqrt{\hat{\lambda}}x)$$

$$X(0) = B = 0$$

$$X(L) = A \sinh(\sqrt{\hat{\lambda}}L) = 0$$

$$\sqrt{\hat{\lambda}}L = 0 \Rightarrow \sqrt{\hat{\lambda}} = 0 \Rightarrow \hat{\lambda} = 0$$

a contradiction, hence $B = 0$

Trivial

Parabolic PDEs: Heat Eq. $u_t = u_{xx}, \quad 0 < x < L$
 $\Delta = 0 \quad t \geq 0$

Dirichlet BCs: $u(0, t) = 0, \quad u(L, t) = 0$

ICs: $u(x, 0) = f(x)$

* Sep of variables: $u(x, t) = X(x)T(t)$

$$u_t = X T' \Rightarrow \frac{T'}{T} = \frac{X''}{X} = -\lambda^2$$

• Non trivial sol^{ns} $\lambda^2 > 0$

$$T' = -\lambda^2 T \Rightarrow T(t) = e^{-\lambda^2 t}$$

$$X(x) = A \cos(\lambda x) + B \sin(\lambda x)$$

$$u(0, t) = u(L, t) = 0$$

$$X(0) = A = 0$$

$$X(L) = B \sin(\lambda L)$$

$$\lambda L = n\pi$$

$$\lambda_n = \frac{n\pi}{L}$$

$$X_n = B \sin\left(\frac{n\pi x}{L}\right) \longrightarrow T_n(t) = d e^{-\left(\frac{n\pi}{L}\right)^2 t}$$

$$u_n(x, t) = b_n e^{-\frac{n\pi}{L} t} \sin\left(\frac{n\pi x}{L}\right)$$

* Most general solⁿ:

$$u(x, t) = \sum_{n=0}^{\infty} d_n e^{-\left(\frac{n\pi}{L}\right)^2 t} \sin\left(\frac{n\pi x}{L}\right)$$

$$u(x, t) = \sum_{n=0}^{\infty} d_n e^{-\left(\frac{n\pi}{L}\right)^2 t} \sin\left(\frac{n\pi x}{L}\right)$$

$$u(x, 0) = f(x) = \sum_{n=0}^{\infty} d_n \sin\left(\frac{n\pi x}{L}\right)$$

Idea: solve for d_n in terms of f :

$$\int_0^L \sin\left(\frac{m\pi x}{L}\right) f(x) dx = \int_0^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$\int_0^L \sin\left(\frac{m\pi x}{L}\right) f(x) dx = \sum_{n=0}^{\infty} d_n \frac{L}{2} \delta_{mn}$$

$$\therefore d_n = \frac{2}{L} \int_0^L \sin\left(\frac{n\pi x}{L}\right) f(x) dx$$

$$u(x, t) = \sum_{n=1}^{\infty} d_n \sin\left(\frac{n\pi x}{L}\right) e^{-\left(\frac{n\pi}{L}\right)^2 t}$$

Hyperbolic PDE: $\Delta > 0$

$u_{tt} = u_{xx}$, (wave eq)

B.C.: $u_x(0, t) = 0$ } Neumann

$u_x(L, t) = 0$ } Dirichlet

{ 2 BC }

{ 2 IC }

→ Zero flow out of the system

I.C.: $u(x,0) = f(x)$

$u_t(x,0) = g(x)$

$u = XT$

$$\frac{T''(t)}{T} = \frac{X''}{X} = \underbrace{-\lambda^2}_{\leq 0}$$

$X'' = -\lambda^2 X$

$X'(0) = 0$

$X = A \sin(\lambda x) + B \cos(\lambda x)$

$X'(L) = 0$

$X' = A \lambda \cos(\lambda x) - B \lambda \sin(\lambda x)$

$X'(0) = A \lambda = 0$

$A = 0$

$X'(L) = -B \lambda \sin(\lambda L) = 0$

$B \neq 0: \lambda L = n\pi \Rightarrow \lambda_n = \frac{n\pi}{L}$

$X_n(x) = B \cos(\frac{n\pi}{L} x)$

$T_n(t) = \hat{A} \cos(\frac{n\pi}{L} t) + \hat{B} \sin(\frac{n\pi}{L} t)$

$\left\{ \begin{array}{l} \lambda^2 = 0: X'' = 0 \Rightarrow X(x) = A + Bx \\ X' = B = 0 \\ X(x) = A_0 \end{array} \right. \quad \left. \begin{array}{l} T_0 = C_0 + D_0 t \\ \lambda_0 = 0 \end{array} \right.$

$u(x,t) = \sum_{n=0}^{\infty} X_n(x) T_n(t) = X_0(x) T_0(t) + \sum_{n=1}^{\infty} X_n(x) T_n(t)$
 $= A_0 (C_0 + D_0 t) + \sum_{n=1}^{\infty} \tilde{B}_n \cos(\frac{n\pi}{L} x) (B_n \cos(\frac{n\pi}{L} t) + C_n \sin(\frac{n\pi}{L} t))$

$u(x,t) = \tilde{A}_0 + \tilde{C}_0 t + \sum_{n=1}^{\infty} \cos(\frac{n\pi}{L} x) (\hat{a}_n \cos(\frac{n\pi}{L} t) + \hat{b}_n \sin(\frac{n\pi}{L} t))$

$\tilde{A}_0 := \frac{1}{2} a_0, \quad \tilde{C}_0 := \frac{1}{2} b_0$

$u(x,t) = \frac{a_0 + b_0 t}{2} + \sum_{n=1}^{\infty} \cos(\frac{n\pi}{L} x) [\hat{a}_n \cos(\frac{n\pi}{L} t) + \hat{b}_n \sin(\frac{n\pi}{L} t)]$

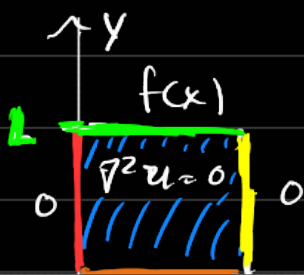
Elliptic PDEs:

$\nabla^2 u = u_{xx} + u_{yy} = 0$

Laplace's Eq

$0 < x < L$

$0 < y < L$



$\int \begin{array}{l} u(0,y) = 0 \\ u(x,0) = 0 \end{array}$

0 L

$$u(L, y) = 0$$

$$u(x, L) = f(x)$$

$$u_t = u_{xx}$$

IC • $u(x, 0) = \begin{cases} 2x & 0 \leq x \leq \frac{1}{2} \\ 2-2x & \frac{1}{2} \leq x \leq 1 \end{cases}$

BC $\frac{\partial u}{\partial x} \Big|_{x=0} = 0, \frac{\partial u}{\partial x} \Big|_{x=1} = 0$

$$u(t, x) = T(t) X(x)$$

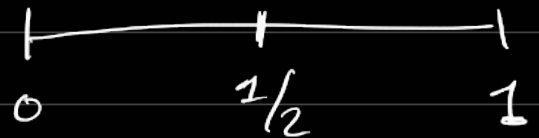
$$\frac{T'}{T} = -\lambda^2 = \frac{X''}{X}$$

$$X'' = -\lambda^2 X \Rightarrow X(x) = A \cos(\lambda x) + B \sin(\lambda x)$$

$$T' = -\lambda^2 T \Rightarrow T(t) = C e^{-\lambda^2 t}$$

$$* \frac{\partial^2 u}{\partial x^2} = -A \lambda \sin(\lambda x) + B \lambda \cos(\lambda x)$$

$$\frac{\partial u}{\partial x} \Big|_{x=0} = B \lambda = 0$$



$$\lambda \neq 0 \Rightarrow B = 0$$

$$* X(x) = A \cos(\lambda x)$$

$$X'(x) = -A \lambda \sin(\lambda x)$$

$$X'(0) = -A \lambda \sin(\lambda) = 0$$

$A \neq 0$: Trivial solution

$\lambda \neq 0$:

$$\Rightarrow \cos(\lambda) = 0$$

$$\lambda_n = \frac{\pi}{2} + n\pi = \pi \left(\frac{2n+1}{2} \right)$$

$$\begin{cases} X_n(x) = A \cos(\lambda_n x) \\ T_n(x) = C e^{-\lambda_n^2 t} \end{cases}$$

$$\lambda < 0$$

$$\lambda = 0:$$

$$T' = 0 \Rightarrow$$

$$T_0(t) = \text{const.} = \tilde{C}$$

$$\ddot{X} = 0$$

$$X_0(x) = Ax + B$$

$$\frac{\partial X}{\partial x} \Big|_{x=0} = \tilde{A} = 0 \quad \Rightarrow \quad \hat{A} = 0$$

$$X_0(x) = \text{const.}$$

